

STEP 1: Create Project Directory

```
mkdir program-2  
cd program-2
```

STEP 2: Create Folder Structure

```
mkdir src
```

STEP 3: Create src/index.js

```
nano src/index.js
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~$ mkdir program-2  
cd program-2  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ mkdir src  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ nano src/index.js  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ cat src/index.js  
const express = require('express');  
const app = express();  
const PORT = 3000;  
  
app.get('/', (req, res) => {  
  res.send('Hello from multi-stage Docker!');  
});  
  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

STEP 4: Create package.json

```
nano package.json
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ npm init -y  
Wrote to /home/1RV24MC087_SSDhanush/program-2/package.json:  
  
{  
  "name": "program-2",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": ""  
}
```

STEP 5: Generate package-lock.json

```
npm install
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ npm install  
  
up to date, audited 1 package in 424ms  
  
found 0 vulnerabilities
```

STEP 6: Create Dockerfile

nano Dockerfile

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ nano Dockerfile
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ cat Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./

RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD ["node", "dist/index.js"]
```

STEP 7: Build Docker Image

docker build -t program-2 .

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker build -t program-2 .
[+] Building 18.3s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 427B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  1.7s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:658d0f63e501824d6c23e06d4bb95c71e7d704537c9d9272f488ac03a3 0.0s
=> [internal] load build context                                   0.2s
=> => transferring context: 2.33MB                                   0.2s
=> CACHED [builder 2/6] WORKDIR /app                               0.0s
=> [builder 3/6] COPY package.json package-lock.json ./           0.1s
=> [builder 4/6] RUN npm install                                   14.8s
=> [builder 5/6] COPY . .                                          0.4s
=> [builder 6/6] RUN npm run build                                 0.5s
=> [stage-1 3/6] COPY --from=builder /app/package.json ./         0.0s
=> [stage-1 4/6] COPY --from=builder /app/package-lock.json ./   0.1s
=> [stage-1 5/6] COPY --from=builder /app/dist ./dist             0.0s
=> [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules 0.1s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.1s
=> => writing image sha256:fea94a53bd8dd62eee7166aeaaadcd130f3a3643b41af99e2708203d2cbb294 0.0s
=> => naming to docker.io/library/program-2                       0.0s
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker run -it -p 3000:3000 program-2
```

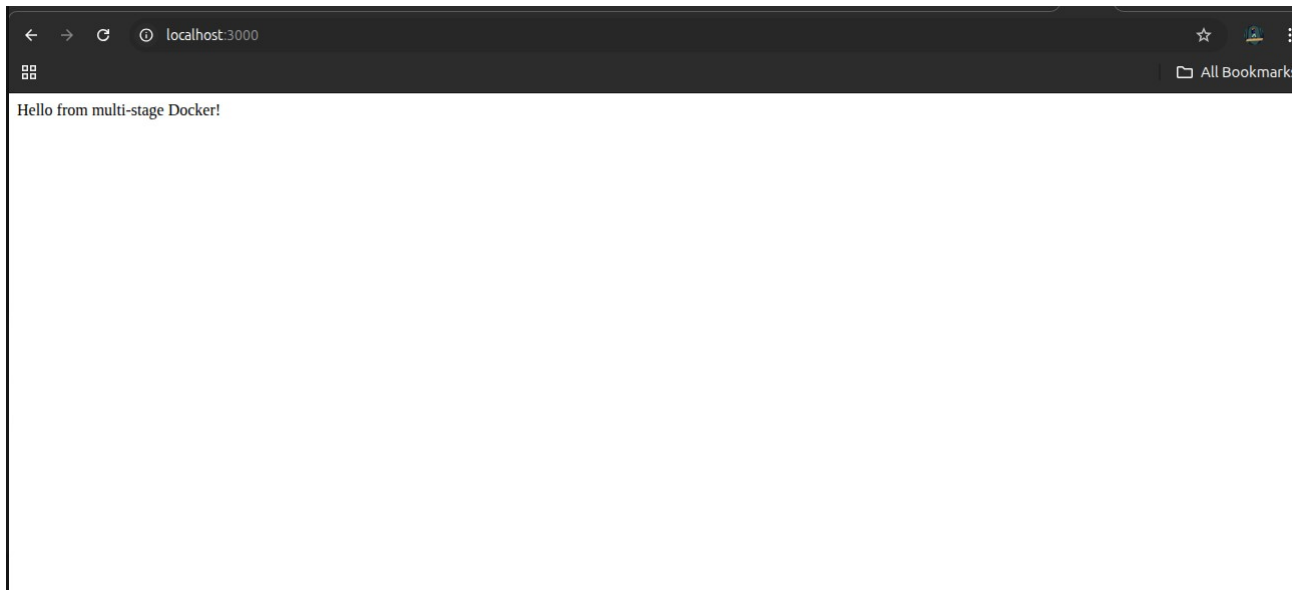
STEP 8: Run Docker Container

docker run -it -p 3000:3000 program-2

```
=> => naming to docker.io/library/program-2                        0.0s
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker run -it -p 3000:3000 program-2
Server running on port 3000
```

STEP 9: Test in Browser

Open browser:



STEP 10: Stop the Container

```
docker ps
```

```
docker stop <container-id>
```

STEP 11: Remove Container & Image

```
docker container ls -a
```

```
docker container rm <container-id>
```

```
docker image rm program-2
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
program-2     latest   fea94a53bd8d   4 minutes ago  137MB
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker ps
CONTAINER ID   IMAGE        COMMAND                  CREATED        STATUS        PORTS
NAMES
347a567a14bc   program-2    "docker-entrypoint.s..." 4 minutes ago  Up 4 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
clever_pare
```

```
docker: Command not found
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker stop 347a567a14bc
347a567a14bc
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker rm 347a567a14bc
347a567a14bc
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker rmi program-2
Untagged: program-2:latest
Deleted: sha256:fea94a53bd8dd62eee7166aeafaadcd130f3a3643b41af99e2708203d2cbb294
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$ docker ps
docker images
CONTAINER ID   IMAGE        COMMAND                  CREATED        STATUS        PORTS        NAMES
REPOSITORY    TAG          IMAGE ID       CREATED        SIZE
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-2$
```